

ESM Credit LOS — Codebase Map

A directory-by-directory guide to the credit module, backend and frontend. All paths relative to repo root.

1. Backend module root: `backend/src/credit/`

```
backend/src/credit/
■■■■ routes/ Express routers (~74 files) – credit.routes.ts is the master
■■■■ controllers/ 69 HTTP controllers (thin; delegate to services)
■■■■ services/ ~108 service files (business logic – the meat)
■■■■ middleware/ 14 dedicated middleware (authz, scoping, flags, MFA, DLP, SOD, audit)
■■■■ adapters/ External integrations (bureau, AI, AV-scanner registry)
■■■■ validators/ Joi request validators
■■■■ queues/ BullMQ queue setup (queues/index.ts)
■■■■ jobs/ Recurring jobs (SLA checker, AML rescreen, collateral insurance, audit retention,
LOO expiry, monitor)
■■■■ schedulers/ node-cron schedulers (looExpiry.scheduler.ts)
■■■■ templates/ Email/PDF templates
■■■■ types/ Shared types & enums (credit.types.ts)
■■■■ constants/ Domain constants (e.g. financialStatementTemplates.ts)
■■■■ utils/ Helpers
■■■■ __tests__/ Tests (also per-directory __tests__/)
```

Routes — the entry points

`routes/credit.routes.ts` is the **master router**. Everything mounts here. Its structure (by comment grouping):

- Global gates: service-API-key AV-callback (unauthenticated) + feature-flags endpoints (before auth), then authenticate + `requireFeatureFlag('credit:module')`, then health.
- `/applications/:applicationId` parent guard (line 242): `applyRmScope()` + `requireApplicationAccess()` — inherited by all nested app-scoped routers.
- Mounted domains (each is a sub-router under `routes/`):
- **Borrower & parties:** `/borrowers` (borrowerProfile, director, fatcaCrs, shareholder, ubo), `/borrowers/duplicate-exceptions`, `/related-party-groups`, `/branches`
- **Documents:** `creditDocumentRoutes` (mounted at root), `/applications/.../ca-memo*`, `/applications/.../ca-memo-versions*`, `/applications/.../approval-pack`
- **Applications:** `/applications` (application, applicationFacility, applicationParty, requestItem, exposureSummary, externalRating, ecl, projection, sensitivityScenario)
- **Approval:** `approvalRoutes` (root), `/approvals`
- **Webhooks:** `/webhooks`
- **Financials:** `/borrowers`, `/financials`, `/scorecards`, `/scorecard-versions`, `/applications`, `/score-runs`
- **Committee:** `/committee`
- **Collateral:** `/applications`, `/collateral`
- **Guarantees:** `/applications`
- **Conditions:** `/applications`, `/conditions`
- **Dashboard:** `/dashboard`; **Reports:** `/reports`
- **Monitoring:** `/applications`, `monitoringItemRoutes`
- **Security:** `/security`
- **Recommendation:** `creditRecommendationRoutes`; **Borrower risk:** `borrowerRiskRoutes`

- **CA Memo Phase 4:** /applications (profitability, walletShare, keyCounterparty, accountUtilisation)
- **Score override:** /score-overrides; **Delegation:** /delegation
- **SLA:** /sla; **DLP:** root; **Disbursement:** /applications; **Pricing:** /applications; **LOO:** /applications; **Rejection:** /applications; **AML rescreen:** root
- **Policy limits:** /policy-limits; **Rule config:** root; **FX rates:** /fx-rates
- **Credit AI:** /applications
- **Deviation:** /deviations; **Consent:** /consent; **STR:** /str; **MFA:** /mfa
- **Lane:** /applications/:id/lane, /applications/:id/tabs
- **SME financial:** /sme; **Comments:** root

> The route file comments are grouped by **sprint / phase / §section** (Sprint 1–5, CA Memo Phases 2–5, §1.x/§2.x/§6.x, P-x). This is the fastest way to understand the module's build history and intent.

Controllers — the HTTP handlers

backend/src/credit/controllers/ (69 files). Naming is domain.controller.ts. Thin: parse/validate request, call a service, shape the response. Notable non-per-domain ones:

- caMemoPdf.controller.ts — generates CA memo PDF.
- approvalPack.controller.ts — builds the approval pack.
- lane.controller.ts — lane determination + tab routing.
- clamav.controller.ts — AV-scanning callback.
- exportToken.controller.ts — DLP export tokens.

Services — business logic (the meat)

backend/src/credit/services/ (~108 files). Key ones for orientation:

- creditApplication.service.ts (1,943 lines) — the state machine (TRANSITIONS), createApplication/updateApplication/transitionApplication, application number generation, audit trail, evidence mapping, clone. **Read this first.**
- stateGuard.util.ts — sub-resource edit/deletion guards by parent state.
- creditRuleEngine.service.ts, creditRuleDefaults.ts, missingDataPolicy.service.ts — rules engine.
- scoring.service.ts, scorecard.service.ts, scoreFactorDefinition.service.ts, ratingBand.service.ts, scoreOverride.service.ts, recalc.service.ts — scoring & risk rating.
- approvalMatrix.service.ts, approvalAction.service.ts, committee.service.ts, committeeEntryGate.ts, approvalPack.service.ts, signoff.service.ts — approval chain.
- creditSla.service.ts, lane.service.ts, dashboard.service.ts, reports.service.ts — ops.
- auditChain.service.ts, dlp.service.ts, exportAudit.service.ts, piiReadLog.service.ts, encryption.service.ts — security/audit.
- credit-ai.service.ts, creditAiCompliance.service.ts — AI-assisted credit.
- borrowerOnboarding.service.ts, borrowerProfile.service.ts, borrowerCreditData.service.ts, borrowerDuplicateException.service.ts, borrowerRisk.service.ts — borrower.
- caMemoPdf.service.ts, htmlToPdf.service.ts — document generation.
- disbursement.service.ts, loo.service.ts, condition.service.ts, rejection.service.ts — downstream lifecycle.

Middleware — cross-cutting

backend/src/credit/middleware/ (14): applicationAccess, assertBorrowerAccess, assertCreditDocumentAccess, autoAudit, consent, dlp, featureFlag, fieldEncryption, mfa, piiReadLog, rmScope, sod. (Plus __tests__.)

Jobs & schedulers

backend/src/credit/jobs/: amlRescreenChecker.ts, auditRetention.job.ts, collateralInsuranceMonitor.job.ts, creditSlaChecker.ts, loanExpiry.job.ts, monitor.job.ts.

backend/src/credit/schedulers/: loanExpiry.scheduler.ts.

backend/src/credit/queues/: index.ts (BullMQ).

2. Backend mount point (outside the module)

backend/src/routes/index.ts:95 → router.use('/credit', creditRoutes) → all endpoints under /api/v1/credit. The API_PREFIX env var is configurable.

3. Prisma schema (backend/prisma/schema.prisma)

- **210 models** total; ~60+ are credit-related (see 02-data-model.md).
- ApplicationState, FacilityType, RiskRating, CommitteeVoteChoice, DisbursementStatus, etc. are Prisma **enums** here — single source of truth. Import from @prisma/client, don't duplicate.
- God-models to treat carefully: CreditApplication (~68 fields / 40+ relations), BorrowerProfile, User (~106 fields / 80+ relations).

4. Frontend

```
frontend/  
■■■ App.tsx React Router — /credit subtree (see 06-frontend-map.md)  
■■■ pages/credit/ Credit pages (dashboard, create, committee, reports, mobile)  
■■■ pages/CreditApplicationList.tsx / CreditApplicationDetail.tsx (root pages dir)  
■■■ src/components/credit/ Credit components, incl. borrower360/ workspace  
■■■ src/services/  
    ■■■ credit.service.ts (3,775 lines) — main credit API client  
    ■■■ credit.types.ts shared types/enums  
    ■■■ creditAi.service.ts AI-assisted credit client
```

See 06-frontend-map.md for the full frontend breakdown.

5. How to navigate a feature end-to-end (worked example)

Take "make an offer on a credit application":

- 1 **Frontend** — application detail workspace → Conditions/Offer tab calls frontend/src/services/credit.service.ts.
- 2 **HTTP** — POST /api/v1/credit/applications/:id/transition (mounted in credit.routes.ts → creditApplicationRoutes).
- 3 **Controller** — creditApplication.controller.ts parses { action: 'make_offer' }, applies requireTransitionPermission.

- 4 **Service** — `creditApplication.service.ts::transitionApplication` → finds transition, runs **CP-fulfilment gate**, guarded `$transaction`, writes audit event.
- 5 **Prisma** — updates `CreditApplication.state` to `OFFER`, increments version.

Follow this same path for any feature: route → controller → service → schema.